

NeuroShift: 10 Reference Papers (Module-wise)

For guide review | Base paper: Chamberland et al., IEEE EMBC 2024 | July 2026

We selected 10 relevant papers. Chamberland EMBC 2024 is the BASE. Each NeuroShift module maps to one or more of these papers. Novelty = gaze selects + EMG confirms + Act/Abstain + FAR for smart-home assistance.

1. Explain to guide (30 seconds)

- Surveyed EMG, vision, gaze, fusion, smart-home assistive, and abstention literature.
- Chose 10 key papers; P1 Chamberland is base (sir already approved direction).
- Built modules from these papers; novelty is the combination for home FAR safety.

2. The 10 papers

#	Paper (short)	Module	Role for NeuroShift
P1	Chamberland EMBC 2024: Visual Scene Understanding for Enhanced EMG Gesture Recognition	BASE + EMG+vision	Base paper; extend gestures -> home intention
P2	Frontiers 2024: Multimodal EMG+vision grasp INTENT inference	Fusion / intention	Supports intent (not only gesture class)
P3	IEEE TNSRE 2024: Hybrid Gaze + FEMG assistive control (SCI)	Gaze select + EMG confirm	Closest pattern: looking selects, muscle confirms
P4	IEEE ICCIT 2023: BlinkGrid gaze control of home appliances	Gaze -> home devices	Webcam gaze for appliances; we add EMG+FAR
P5	Hybrid smart home SSVEP+EMG (confirm / anti-idle errors)	Confirm + safety	EMG confirms selection; reduces false ops
P6	Scheme TBME 2013: Confidence-based rejection myoelectric	Act/Abstain	Classic reject option when unsure
P7	arXiv 2025 Sagacity: EMG multimodal smart-home for	Smart-home EMG domain	Assistive home IoT motivation
P8	MediaPipe Face Mesh (arXiv 1907.06724)	Head/face landmarks	Phase-1 head direction from webcam
P9	YOLO (as in P1) / RT-DETR CVPR 2024	Object detection	Detect lamp/fan/plug in real time
P10	Chamberland TBioCAS 2023 wearable HD-EMG (lineage)	Wearable EMG sensing	Wearable EMG valid; we use sparse MyoWare

3. Module-wise mapping (what sir asked)

Our module	Does what	Main papers	We improve by
Appliance detection	Find devices in camera	P1, P9	Home appliances, not only grasp objects
Gaze / head select	Which device user looks at	P3, P4, P8	Select for home; Phase1 head, Phase2 eyes
EMG confirm	Confirm real toggle intent	P3, P5, P10	Sparse MyoWare confirm, not gesture dictionary
Multimodal fusion	Combine gaze target + EMG	P1, P2, P3	Output = Act/Abstain for home
Act / Abstain	No action if unsure	P5, P6	Home False Actuation Rate (FAR)
Smart-home IoT	ESP32 + relay control	P7	Indian wired-home relay retrofit
Overall system	Full pipeline	P1 BASE	Gaze-select + EMG-confirm home intention

4. Flow from papers to our idea

P9/P1 detect appliances -> P8/P4 gaze selects -> P10/P3 EMG confirms -> P2/P1 fuse -> P5/P6 abstain if unsure -> P7 home assistive goal.

BASE = P1 Chamberland. Novelty = combination for safer home intention (FAR).

5. Key DOIs / IDs

- P1 BASE DOI: 10.1109/EMBC53108.2024.10782354
- P2 DOI: 10.3389/frobt.2024.1312554
- P3 DOI: 10.1109/tnsre.2024.3443073
- P4 DOI: 10.1109/iccit60459.2023.10441588
- P6 DOI: 10.1109/TBME.2013.2238939
- P7 arXiv: 2507.19479
- P8 arXiv: 1907.06724
- P9 RT-DETR arXiv: 2304.08069 / CVPR 2024
- P10 DOI: 10.1109/TBCAS.2023.3314053

6. Novelty vs these 10

- Not claiming first EMG+camera or first gaze home control.
- Claiming: gaze-select + sparse EMG-confirm + Act/Abstain for home devices, measured by FAR.
- Extends P1 from prosthesis gesture rescue to assistive smart-home intention.

Primary base paper: Chamberland et al., IEEE EMBC 2024. Supporting module references: P2-P10. Total core set for guide: 10 papers.